Finite Sector Extraction and Persistence Obligations

From Bounded Algebraic Prototypes to the Open O-SECTOR Bridge

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Abstract

This paper isolates the sector-extraction frontier between the flagship kernel theorem package and any later field-theoretic bridge. Its positive content is finite and bounded: certified extraction procedures can produce algebraic prototypes, interaction-type quotients, and representation-theoretic sectors on named regimes, and in some cases can verify closure or quotient-stability properties within those regimes. Its negative content is equally important: none of this yet proves persistence under enlargement, canonicality across extraction routes, or projection from a distinguished finite sector to a continuum-relevant physical sector. The paper therefore formulates O-SECTOR as the exact theorem stack separating finite extracted algebra from scaling-stable mathematical meaning.

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1 Introduction

The dream-paper suite cannot move honestly from the flagship kernel paper directly to Yang–Mills or Navier–Stokes rhetoric. There is an intermediate burden. That burden is not primarily analytic PDE control. It is the problem of proving that finite extracted algebra survives scaling in a canonical way.

The project already contains real finite algebraic structure: frozen-regime interaction-type quotients, screened-sector data, bounded transfer objects, finite group-algebra decompositions, and conditional identification of distinguished non-abelian sectors in named finite settings. These are not empty gestures. But they remain bounded results unless one proves that the distinguished algebraic content persists under admissible enlargement and does not depend on accidental finite choices.

This is the purpose of the present paper. It turns the vague phrase "sector emergence" into a sharply bounded finite program plus an explicit bridge obligation. That bridge obligation is O-SECTOR.

2 What counts as a sector in this paper

The word "sector" is used at more than one strength in the project. The first task is therefore terminological discipline.

Definition 2.1 (Finite extracted sector)**.** A *finite extracted sector* is an algebraic object produced from a named bounded regime by a licensed extraction procedure, together with only those properties that have actually been proved or computationally certified on that regime.

Definition 2.2 (Persistent sector)**.** A *persistent sector* is an algebraic object equipped with a canonical identification across an enlargement family of regimes, together with a proof that its defining structure survives the enlargement procedure in a mathematically stable sense.

Remark 2.3. A finite extracted sector may be informative without being persistent. The distinction is not cosmetic. It is the main referee barrier between a serious finite algebraic result and an overclaimed scaling narrative.

3 Finite sources of algebraic sector data

The current project provides several finite sources from which sector candidates arise.

3.1 Interaction-type quotients

At the frozen-regime level, one can classify admissible interaction types by terminal quotient data. For odd prime K_0, the full interaction-type quotient has the form

$$|Q_{\text{hier}}| = K_0 + 2,$$

with diagonal and mixed classes separated by layered invariants such as state-space dimension, spectral data, and slope-type invariants. This is sector information in a finite combinatorial sense: it isolates algebraically distinct classes that cannot be collapsed by the admissible finite symmetry group.

3.2 Finite representation sectors

On the finite algebra side, an extracted non-abelian sector may arise from a representation-theoretic decomposition. In the project's conditional gauge-algebra supplement, the extended finite group algebra admits an Artin–Wedderburn decomposition whose non-abelian blocks are matrix algebras, and the Schr"odinger representation isolates a distinguished irreducible summand carrying the non-abelian commutator algebra.

3.3 Screened or selected sectors

Other parts of the project define screened-sector or selected-sector data by finite admissibility and diagnostic criteria. These are again meaningful finite objects, but they only become persistent sectors after a separate stability proof across enlargement.

4 Licensed finite claims

The finite extraction program can honestly establish at least the following types of result.

(i) existence of a candidate algebraic object on a named finite regime;

(ii) internal closure of an extracted operation or bracket on that regime;

(iii) dimension, rank, or decomposition data for the extracted object;

(iv) route-agreement between independently implemented extractions on the same regime;

(v) bounded quotient-stability statements internal to a fixed finite symmetry class.

These are already substantial. They reduce interpretive vagueness and create concrete obligations for any later bridge.

5 A bounded extraction schema

To avoid sector rhetoric becoming informal, we package the extraction procedure explicitly.

Definition 5.1 (Extraction schema)**.** An *extraction schema* consists of:

(a) a named finite regime R;

(b) a finite data object $D(R)$ certified or proved on R;

(c) an extraction map $\mathcal{E}_R$ assigning to $D(R)$ an algebraic output $A(R)$;

(d) a verification package $V(R)$ establishing the bounded properties claimed for $A(R)$.

Proposition 5.2 (Bounded soundness principle)**.** *Let $(R, D(R), \mathcal{E}_R, V(R))$ be an extraction schema. If $V(R)$ verifies a property P of $A(R)$, then one is licensed to claim only that P holds for the extracted algebraic output on the named regime R.*

Proof. This is a scope principle rather than a deep theorem. The verification package $V(R)$ is tied to the named regime and does not quantify over larger regimes or continuum limits. Therefore the conclusion inherits exactly that scope and no more. □

Remark 5.3. This proposition is intentionally modest. Its purpose is anti-smuggling: a verified finite bracket or decomposition is not yet a scaling theorem.

6 Route agreement and bounded canonicality

One place where finite extraction can become stronger is through agreement of multiple extraction routes.

Definition 6.1 (Route agreement)**.** Two extraction schemas on the same finite regime R are said to *agree on R* if there is an explicitly verified isomorphism between their extracted outputs that preserves the finite structure under discussion.

Proposition 6.2 (Bounded canonicality from route agreement)**.** *If two or more licensed extraction schemas agree on the same regime R, then the extracted algebraic object acquires a bounded form of canonicality on R: it is independent of the chosen extraction route up to the verified isomorphism.*

Proof. By hypothesis, the outputs are identified by a verified structure-preserving isomorphism. Therefore any claim about the extracted object that is invariant under the relevant notion of isomorphism is route-independent on that specific regime. □

Remark 6.3. This is useful but still bounded. Route-independence on one window is not the same as persistence across a directed enlargement family.

7 A concrete finite prototype: quotient classes to non-abelian blocks

The project's finite algebra program is already rich enough to illustrate the general distinction.

7.1 Frozen interaction classes

In the frozen odd-prime regime, the interaction quotient distinguishes diagonal classes from mixed classes and prevents collapse by the admissible symmetry action. This is a clean finite classification theorem. It gives named sector candidates and a terminal quotient internal to the fixed regime.

7.2 Finite non-abelian representation sector

Under the conditional generator extension studied in the gauge-algebra supplement, the finite group algebra decomposes into abelian summands and matrix blocks. The distinguished non-abelian block carries the commutator algebra $\mathfrak{sl}(K_0, \mathbb{C})$, with the $K_0 = 7$ case yielding the finite prototype A_6.

7.3 What this does and does not buy

This is exactly the sort of finite extracted algebra that matters. It gives a concrete, computable non-abelian sector and identifies a distinguished irreducible block up to the stated bounded hypotheses. But even this strong result does not by itself identify that block as the physical gauge sector, nor prove that the block persists canonically once the finite extraction regime is enlarged or replaced by a continuum bridge.

8 The exact O-SECTOR burden

We can now state the bridge obligation precisely.

Definition 8.1 (O-SECTOR)**.** The obligation **O-SECTOR** is the theorem stack required to upgrade a finite extracted sector to a persistent sector with scaling-stable meaning.

O-SECTOR contains at least four logically distinct sub-obligations.

(S1) **Persistence under enlargement.** For an admissible enlargement family $R_1 \subseteq R_2 \subseteq \cdots$, one must prove that the extracted sector survives, either literally or through canonical identification maps.

(S2) **Coherence of transition maps.** If sectors are transported from R_n to R_{n+1}, the transport maps must compose coherently and preserve the claimed algebraic structure.

(S3) **Canonicality across extraction choices.** Distinct legitimate extraction procedures must converge to the same sector data in the enlargement regime, not merely on isolated windows.

(S4) **Stability of interpretation.** The reason this object deserves to be called a sector must survive scaling: it cannot dissolve into a finite artifact or depend on a frozen-regime accident.

Remark 8.2. The fourth clause is where many ambitious programs fail. A distinguished finite block may be mathematically real and still fail to encode any stable large-scale meaning.

9 Projection is stronger than persistence

The current project materials show that even when a distinguished finite non-abelian block is isolated, one may still need a further projection theorem before obtaining the intended target algebra.

In the gauge-algebra supplement, the finite prototype can reach a distinguished A_6-type algebra in the $K_0 = 7$ case under an explicit finite-sector identification. But the desired projection to the smaller target factor is still left open and is assigned to the E-CONT bridge. This shows that O-SECTOR is not merely "show the algebra survives." It also includes the problem of proving that the correct sub-sector or projected factor remains canonically selected.

Definition 9.1 (Projection obligation)**.** A *projection obligation* is the requirement to prove that a distinguished finite sector or finite algebraic factor canonically projects to the intended target sector in the enlarged or bridged regime.

Remark 9.2. Projection obligations are especially dangerous because they are easy to smuggle rhetorically. A finite algebra may contain the desired target as a sub-factor while still providing no theorem that selects that factor canonically.

10 Where the current project stands

The proper status description is neither pessimistic nor triumphalist. It is stratified.

10.1 What appears genuinely in hand

The finite program appears to contain the following substantive components.

(i) terminal interaction-type quotients on named frozen regimes;

(ii) bounded separation invariants distinguishing finite sector classes;

(iii) finite algebra decompositions into abelian and non-abelian blocks in certain extended generator settings;

(iv) bounded quotient-stability statements identifying a distinguished finite irreducible block under the declared finite symmetry action.

10.2 What remains open

The following stronger claims remain unlicensed without O-SECTOR closure.

(i) persistence of the extracted sector under enlargement of the regime;

(ii) independence from frozen-window or chosen-generator accidents;

(iii) canonical projection from a distinguished finite block to the intended target factor;

(iv) continuum-relevant interpretation of the persistent sector.

11 Why Paper E must exist before Paper F

Paper F, the Yang–Mills bridge paper, will inevitably need a sector statement of the form "the relevant non-abelian algebra survives with stable identity in the regime to which the field-theoretic bridge applies." Without that statement, a Yang–Mills bridge would rest on a finite algebraic prototype whose persistence has not been justified.

Thus Paper E is not optional packaging. It is the hinge manuscript. It converts finite sector extraction from a computational or algebraic curiosity into a disciplined research frontier with a named theorem burden.

12 Status ledger

Label	Claim	Status
E1	Finite interaction-type quotients on named regimes	Discharged / finite
E2	Finite separation invariants for sector classes	Discharged / finite
E3	Finite algebraic decompositions with distinguished non-abelian blocks	Discharged / finite or conditional-on-regime
E4	Bounded quotient-stability of the distinguished finite block	Partially discharged
E5	Persistence of extracted sector under enlargement	Open
E6	Coherent transition maps across enlarging regimes	Open
E7	Canonical projection to the intended target factor	Open
E8	Continuum-relevant stable interpretation	Open

13 The correct positive program

The right next steps are not to overstate the present algebra, but to harden it.

(1) Build extraction schemas that are explicitly comparable across window size and generator class.

(2) Prove route-agreement on overlapping finite regimes.

(3) Identify monotone invariants or transport maps that survive enlargement.

(4) Separate quotient-stability from projection: do not treat a distinguished finite block as the target sector unless the projection theorem exists.

(5) Formulate continuum claims only after persistence and projection are both controlled.

14 Conclusion

This paper gives the dream-paper suite its needed middle layer. The project does not jump directly from finite certificates to field theory. Instead it passes through a mathematically disciplined sector-extraction paper that preserves the real finite progress while making the remaining burden exact.

The core message is simple. Finite extracted algebra is already meaningful. But persistent sector meaning is a stronger theorem, and that stronger theorem is O-SECTOR. Until O-SECTOR and the associated projection obligations are proved, the finite algebra remains a bounded prototype rather than a completed bridge.